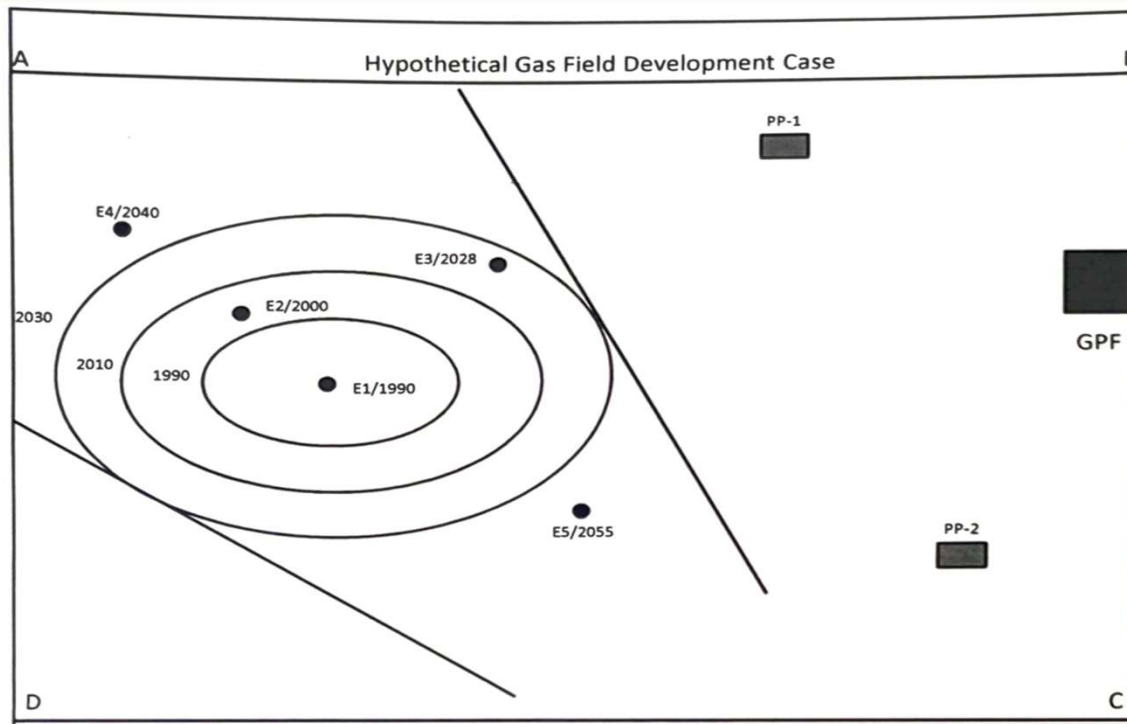


CASE STUDY: DEVELOPMENT OF GAS RESERVOIR USING CLASSICAL METHOD



- All depths in meters.
- GSC: 2030 m (Gas Shale Contact).
- E1, E2 & E3: Gas wells.
- E4 & E5: Dry wells.
- PP-1 & PP-2: Processing facilities.
- GPF: Gas processing facility.
- Distance between A & B: 12 km (Corner points of rectangle).
- Distance between B & C: 10 km (Corner points of rectangle).

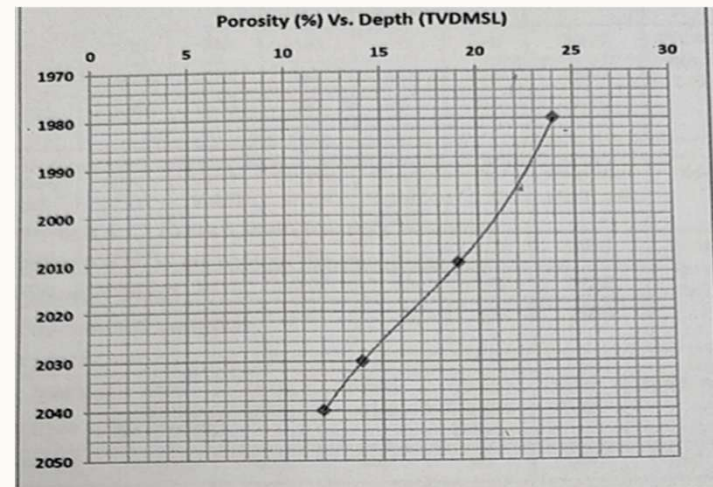
Hypothetical Exploratory well test results

Well	Choke	FTHP, psi	Gas, m ³ /d	Cond. bbl/d	Water, m ³ /d	Cond. Gravity (°API)	FTHT (°F)	Stabilized FBHP, psi	Remarks
E1 Object-1 (1982-1988)	24/64"	3200	230000	900	Nil	50	120	3925	P*: 4000 psi, Res. Temp: 240 deg F
	32/64"	3000	315000	1000	Nil	51	135	3825	
	40/64"	2775	415000	1400	Nil	50	150	3625	

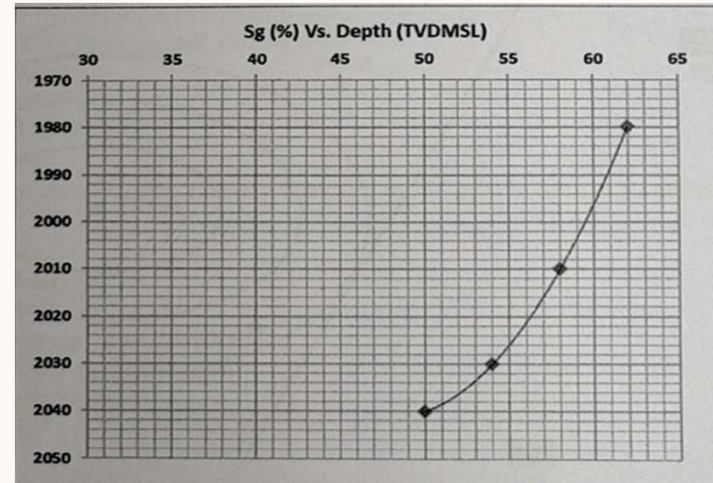
Well	Choke	FTHP, psi	Gas, m ³ /d	Cond. bbl/d	Water, m ³ /d	Cond. Gravity (°API)	FTHT (°F)	Stabilized FBHP, psi	Remarks
E2 Object-1 (2012-2016)	24/64"	3250	228000	950	Nil	50	133	3975	P*: 4050 psi Res. Temp: 242 deg F
	32/64"	3025	300000	1250	Nil	50	148	3825	
	40/64"	2825	375000	1550	Nil	49	165	3650	

Well	Choke	FTHP, psi	Gas, m ³ /d	Cond. bbl/d	Water, m ³ /d	Cond. Gravity (°API)	FTHT (°F)	FBHP, psi
E3 Object-1 (2028-2032)	24/64"	2950	125000	850	Nil	50	140	NM
	32/64"	2750	190000	1250	Nil	50	155	NM
	40/64"	2600	240000	1450	Nil	49	170	NM

Exploratory Well Test results



Porosity vs Depth curve



Gas saturation vs Depth curve

1) ESTIMATION OF GIIP

STEP 1:

Calculated **Area** (using grid method) and **volume** (using pyramidal method) of each region in the contour map.

STEP 2:

Calculated **Porosity** and **Saturation** by taking the arithmetic mean at the top and bounds of the corresponding area.

STEP 3:

Obtained **Gas Deviation Factor** (z) by calculating adjusted P_{pr} and T_{pr} and using the **Standing-Katz chart**. $Z = 0.965$

STEP 4:

Calculated **Gas formation volume factor** (B_g) by substituting z , T and P .

$$B_g = 0.004774 \text{ m}^3/\text{SCM}$$

STEP 5:

Calculated **GIIP** using the volumetric equation by substituting Volume, porosity, saturation and B_g . **GIIP = 11.088 BCM (50% GIIP = 5.544 BCM)**

2) ESTIMATION OF AVERAGE WELL DELIVERABILITY

STEP 1:

Convert the produced condensate to its equivalent gas volume using the provided API gravity and add it to the measured gas production to obtain the **total reservoir gas production**.

STEP 2:

Plotting the Rawlins-Schellhardt back-pressure equation with data from two wells and found AOF for both wells.

$$\text{AOF (well 1)} = 837260.0276 \text{ m}^3/\text{day} \quad \text{AOF(well 2)} = 649397.032 \text{ m}^3/\text{day}$$

STEP 3:

Calculated deliverability, i.e., 25% of AOF, for both wells.

$$\text{For well 1: } 0.21 \times 10^6 \text{ m}^3/\text{d}$$

$$\text{for Well 2: } 0.162 \times 10^6 \text{ m}^3/\text{d}$$

3) CALCULATION OF SBHP AND FBHP

STEP 1:

A **10%** annual exponential decline rate was applied to the initial gas production to estimate the yearly gas production.

STEP 2:

The annual condensate production was calculated using a constant **GCR = 1665.67 m³ gas/m³ condensate**.

STEP 3:

The cumulative gas production and Recovery Factor (RF) **were determined** by summing the annual gas production and dividing it by Original Gas in place.

STEP 4:

The Static Bottom-Hole Pressure (**SBHP**) was estimated by reducing the initial reservoir pressure in proportion to the calculated Recovery Factor.

STEP 5:

The Flowing Bottom-Hole Pressure (**FBHP**) was calculated using a 5% initial drawdown, increasing by 10% annually.

4) CALCULATION OF FTHP

STEP 1:

The **FBHP**, gas production rate, and fluid properties were defined for each year of the 18-year production period.

STEP 2:

B_g, in-situ fluid densities, no-slip liquid holdup, and mixture velocity are calculated.

STEP 3:

Using **Beggs and Brill correlation**, flow regime and liquid holdup are found (up to **12 years**, the flow regime is **intermittent** and **13-18 years** it is **transition**) for both vertical and deviated wells.

STEP 4:

The **hydrostatic and frictional pressure losses** were calculated for both the vertical and deviated wells.

STEP 5:

The **FTHP** was **obtained** by subtracting the total pressure losses from the FBHP, and the production life was evaluated using a minimum surface pressure constraint of **15 kg/cm²**.